

AI-POWERED RESEARCH ASSISTANT USING RAG

Academic Research & Literature Discovery Assistant

Prepared by

Sultanah Al-Mutairi

Ashwaq Ibrahim Al-hulwah

Abrar Al-Furaiddi

Python

RAG

ChromaDB

Embeddings

LLM

Gradio

The Problem & Our Solution



The Problem

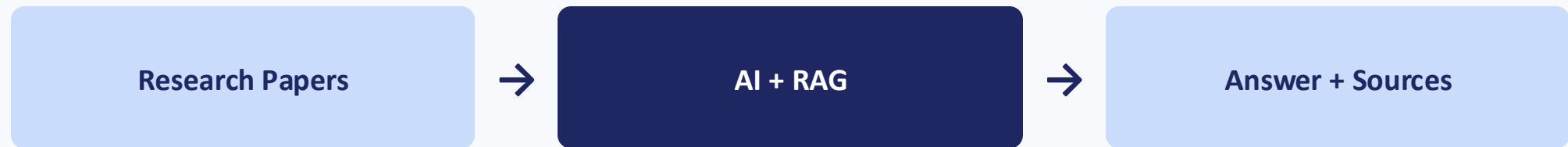
- X Large numbers of academic papers
- X Time-consuming manual research
- X Difficulty finding relevant information
- X Different document formats
- X Risk of unsupported answers



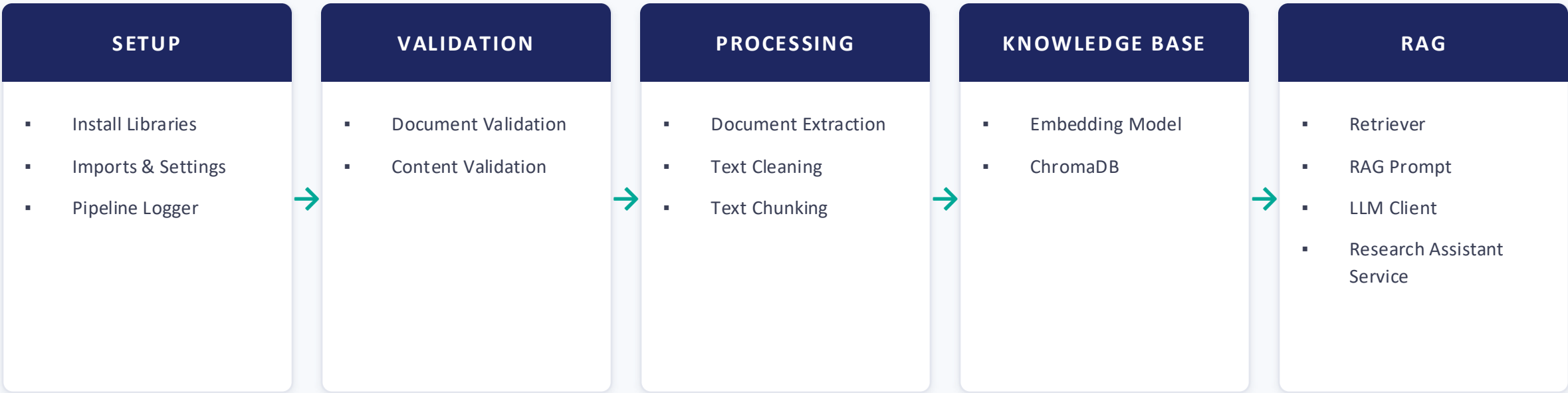
Our Solution

An AI-powered research assistant that:

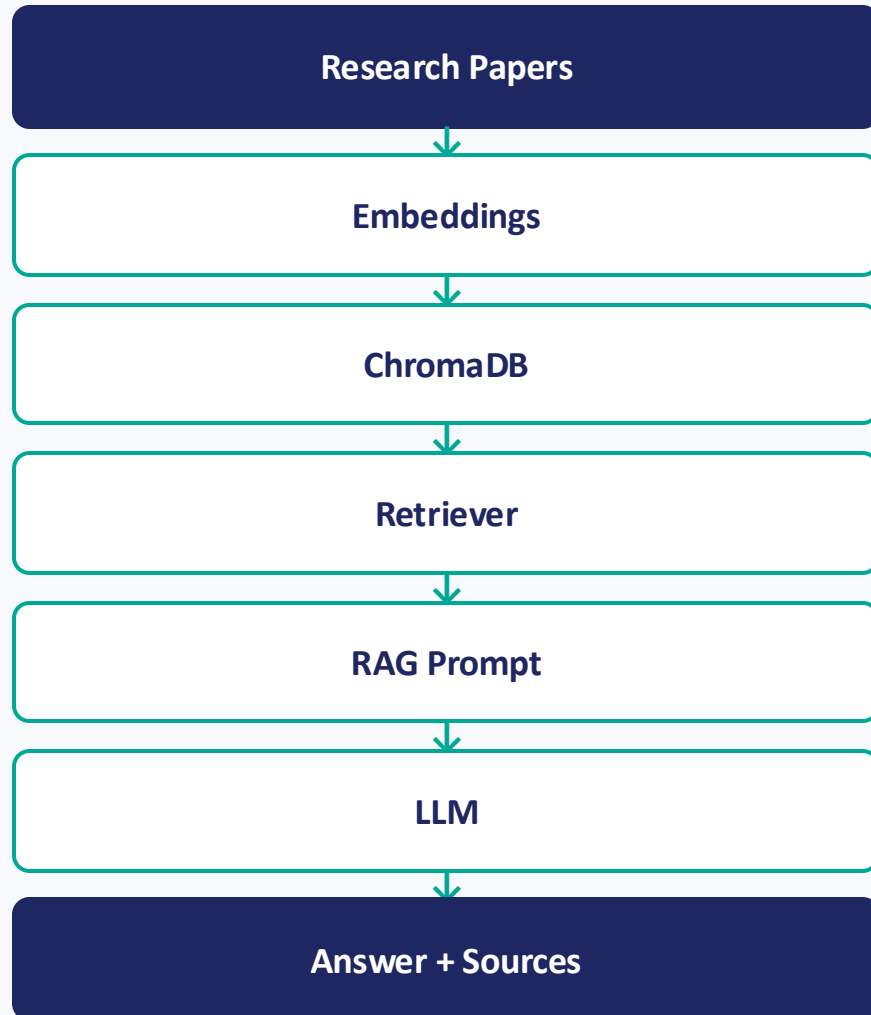
- ✓ Processes academic documents
- ✓ Searches by semantic meaning
- ✓ Retrieves relevant research content
- ✓ Generates source-grounded answers



Project Development Pipeline



How RAG Works

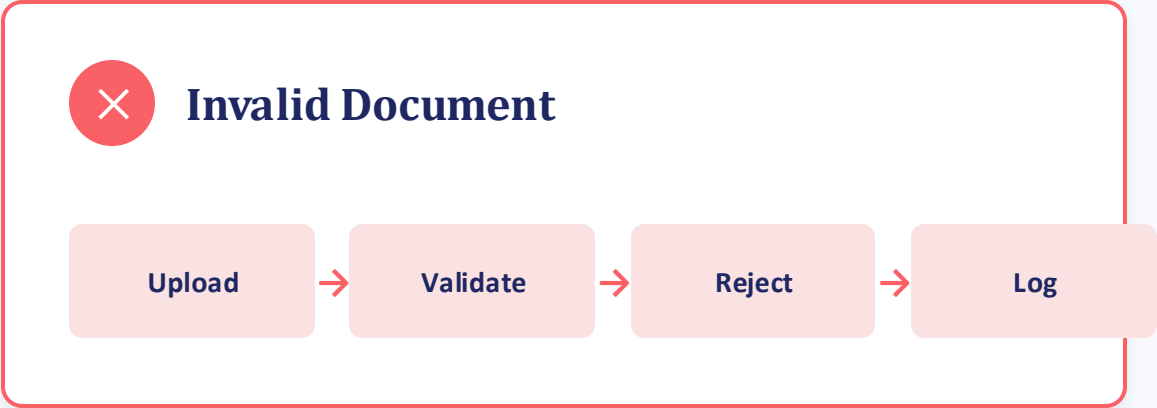
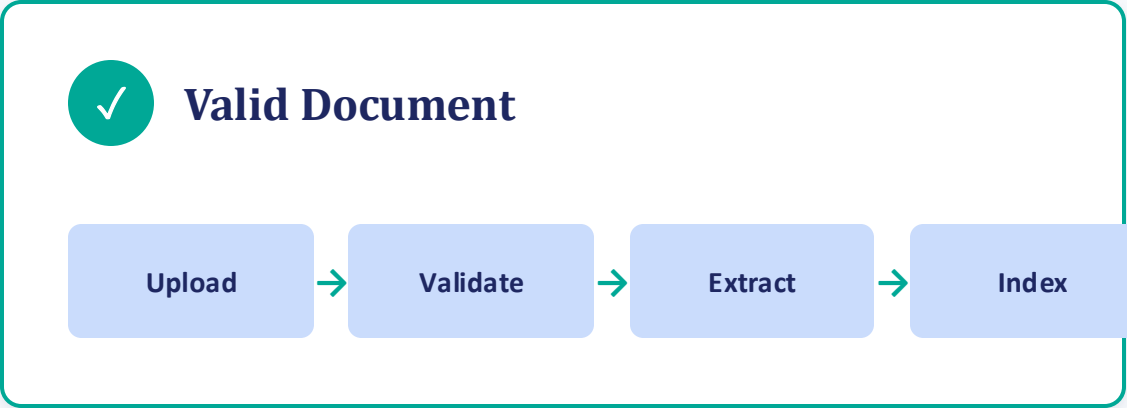


User Question

Retrieval + Context + Generation

- Documents are embedded and stored as vectors in ChromaDB
- A user question retrieves the most relevant chunks
- Retrieved chunks build a grounded RAG prompt
- The LLM generates an answer citing real sources

Document Validation & User Interface



Gradio UI

The user can:

- Upload research documents
- Ask research questions
- View generated answers
- View relevant sources

System Demonstration

1

Upload

valid.pdf

2

Validate

 Validation Passed

3

Index

Chunks → Embeddings → ChromaDB

4

Ask a Question

“What are the main findings of this research?”

5

Result

 AI answer ·  Relevant sources

Conclusion & Future Improvements

The project combines: Python + RAG + Embeddings + ChromaDB + LLM + Gradio

Key Benefits

- ✓ Faster research
- ✓ Semantic search
- ✓ Source-grounded answers
- ✓ Automated document processing
- ✓ Document validation

Future Improvements

- 1 Arabic language support
- 2 Hybrid Search
- 3 Reranking
- 4 Advanced metadata extraction
- 5 Research paper recommendations